

AKI IN CIRRHOSIS — Clinical One-Pager

Differentiate HRS from other AKI · ICA 2015 / AASLD 2021 · terlipressin FDA 2022
· IM resident reference

1 · DIFFERENTIAL — MOSTLY PRE-RENAL

Category	%	Causes
Pre-renal	60–70	Vol-responsive: GIB, over-diuresis, GI losses. Not vol-responsive: HRS , cardiorenal. Also sepsis (SBP, PNA, UTI), meds
Intrarenal	30–40	ATN, AIN, GN
Post-renal	<1	Obstruction, neurogenic bladder

Always, early: diagnostic **paracentesis** (SBP can be painless) + rule out **GIB**. AKI raises mortality 2–6× in cirrhosis.

2 · PATHOPHYS — A VASODILATORY STATE

↓ Liver metabolism of vasodilators → **splanchnic vasodilation** → splanchnic pooling → ↓ **effective arterial volume (EAV)**.

Compensated	Heart ↑CO to defend perfusion
Decompensated	CO can't compensate → ↓BP → RAAS / ADH / sympathetic → renal vasoconstriction + Na/H ₂ O retention → hyponatremia + pre-renal AKI

HRS = the kidney squeezed by systemic vasodilation, not intrinsic renal disease.

3 · HRS-AKI — DIAGNOSIS (ICA)

- Cirrhosis with ascites + AKI (ICA-AKI: ↑Cr ≥0.3 mg/dL in 48h or ≥50% in 7d)
- NO improvement after diuretic withdrawal + **albumin 1 g/kg/d (max 100 g) ×48h**
- No shock; no nephrotoxins; no structural dz (bland UA, no proteinuria/hematuria, normal renal US)

Terminology update: old **Type 1** → **HRS-AKI** (fixed Cr>2.5 threshold dropped); **Type 2** → **HRS-NAKI** (often refractory ascites). ICA 2015 / AASLD 2021.

3b · HRS-AKI — TREATMENT

Albumin	Expand EAV (up to ~50 g/day)
Vasoconstrictor	Terlipressin (FDA 2022) or norepinephrine (ICU); midodrine 5–7.5 mg TID + octreotide 100 µg SQ TID + albumin = older ward regimen. Goal ↑MAP >15
TIPS	Delayed renal benefit; most too ill to qualify
HD	Bridge to transplant
Transplant	Only definitive Rx (±combined liver-kidney)

Terlipressin — avoid: ACLF grade 3, pre-existing hypoxia, or SCr >5 mg/dL (respiratory-failure signal; benefit unlikely).

4 · SBP — DON'T MISS IT

Ascitic ≥**250 PMN/µL** = **SBP** → cefotaxime/ceftriaxone ×5d (FQ if PCN-allergic).

Albumin in SBP: 1.5 g/kg day 1 + 1 g/kg day 3 — reduces HRS & mortality; greatest benefit if **Tbili >4**, **Cr >1**, or **BUN >30** (Sort, NEJM 1999).

URINE INDICES & PITFALLS

- On diuretics → use **FeUrea** (<35% = pre-renal); FeNa is falsely high
- Low urine Na / FeNa can occur even in ATN (RAAS reclaims Na); granular casts also seen w/ severe hyperbilirubinemia — both unreliable for ATN in cirrhosis
- AIN clue:** WBC casts + NSAID use → stop the drug

ATTENDING DOTPHRASE

#AKI in cirrhosis: Cr * (baseline *), AKI stage * [ICA-AKI: ↑Cr ≥0.3 in 48h or ≥50% in 7d]. 2/2 * [pre-renal (hypovol/GIB/over-diuresis/sepsis-SBP/HRS-AKI/cardiorenal), intrarenal (ATN/AIN/GN), post-renal (rare)]. decomp cirrhosis w/ * [ascites/HE/varices/jaundice].
- w/u: meds (diuretics/NSAIDs/contrast), UA+micro, urine lytes [FeUrea <35% pre-renal on diuretics], renal US, **DIAGNOSTIC PARACENTESIS** (≥250 PMN=SBP), r/o GIB
- HRS-AKI dx: cirrhosis+ascites+AKI + NO response to diuretic hold + albumin 1 g/kg/d ×48h + no shock/nephrotox/structural dz
- tx: albumin + vasoconstrictor * [terlipressin (FDA 2022; avoid ACLF-3/hypoxia/SCr>5) / norepi ICU / midodrine+octreotide older] → MAP>15; TIPS select; HD bridge; **DEFINITIVE** = liver transplant (±liver-kidney)
- SBP → cefotaxime/ceftriaxone ×5d + albumin 1.5 g/kg d1 + 1 g/kg d3 (esp Tbili>4/Cr>1/BUN>30)
- AIN (WBC casts+NSAID) → stop drug; cautious volume challenge
- monitoring: strict I/O + daily wt, hold nephrotoxins/NSAIDs/diuretics, renally dose, daily BMP, HE care
- labs: BMP, CBC, LFTs/INR, UA+micro, urine Na/Cr/urea, ascitic cell count+cx, lactate, ±RUQ US Doppler (PVT)